



# Igor Slazyk

ML Research Engineer

## Portfolio

[islazykv.github.io](https://islazykv.github.io)

## LinkedIn

[linkedin.com/in/igor-slazyk](https://linkedin.com/in/igor-slazyk)

## GitHub

[github.com/islazykv](https://github.com/islazykv)

## Personal Information

Nationality: Polish  
Date of Birth: 03.11.1992  
Address: Sandnes, Norway  
Phone Number: [REDACTED]  
E-mail: [islazykv@gmail.com](mailto:islazykv@gmail.com)

## Languages

English: C1  
Norwegian: A2  
Italian: A2  
Polish: Native

## Interests

Popular Science Literature  
IT / Technology  
Gaming  
Cycling  
Hiking

## Summary

Machine Learning Research Engineer with 8 years of experience in data science, including 4 years focused on machine learning. Experienced in building scalable data pipelines and ML systems on large and complex datasets. Driven to work at the intersection of data engineering and ML, designing pipeline infrastructure with a direct understanding of what models actually need from data.

## Education

- **Ph.D. in Particle Physics** 10/2017 – 07/2021  
University of Ferrara Ferrara, Italy  
Institute of Nuclear Physics Krakow, Poland
- **M.Sc. in Applied Physics** 02/2016 – 05/2017  
Krakow University of Technology Krakow, Poland
- **B.Sc. in Applied Physics** 09/2012 – 02/2016  
Krakow University of Technology Krakow, Poland
- **B.Sc. in Civil Engineering** 09/2011 – 02/2015  
Krakow University of Technology Krakow, Poland

## Experience

- **Postdoctoral Researcher** 01/2022 – 05/2026  
Høgskulen på Vestlandet Bergen, Norway  
CERN · ATLAS Collaboration Geneva, Switzerland
  - Built end-to-end, multi-stage data and ML pipelines processing datasets of 10M+ events, from raw ingestion to model-ready outputs
  - Trained and evaluated ML models (XGBoost, PyTorch) for classification, regression and anomaly detection, tracked with MLflow and W&B
  - Developed data transformation software within a distributed processing framework (Athena) operating at petabyte scale
  - Designed scalable ETL/ELT workflows for data preparation and feature engineering
  - Optimized pipeline performance and refactored legacy code in a 4M-line collaborative codebase
  - Implemented data validation and quality checks across pipeline stages
  - Established automated testing and CI/CD pipelines for reliable, reproducible data and ML workflows
  - Communicated analytical results to cross-functional teams of engineers, analysts and domain experts
- **Doctoral Researcher** 10/2017 – 07/2021  
University of Ferrara Ferrara, Italy  
CERN · LHCb Collaboration Geneva, Switzerland
  - Built an automated data quality monitoring system (C++/Python/LabVIEW) for continuous QA of measurement outputs
  - Developed data processing scripts and tooling for structured measurement pipelines
  - Analyzed QA output data to detect anomalies and failure patterns across large datasets
  - Explored ML-based anomaly detection for automated performance classification
- **Junior Front-End Developer** 10/2016 – 09/2017  
Freelance Krakow, Poland
  - Built responsive websites using React.js
  - Collaborated with back-end developers on feature integration
- **R&D Physics Intern** 04/2016 – 09/2016  
Camlin Technologies Zurich, Switzerland
  - Built data collection and processing workflows for semiconductor measurement data
  - Analyzed experimental measurement data to extract and compare material properties
  - Collaborated with software developers and physicists on shared analysis workflows

| Selected Projects                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                           |
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| <ul style="list-style-type: none"> <li><a href="#">repository</a></li> <li><a href="#">notebook showcase</a></li> <li><a href="#">publication</a></li> </ul>                             | <b>Tau Supersymmetry Search</b><br>Boosted Decision Trees   Neural Networks <ul style="list-style-type: none"> <li>Developed a supervised ML pipeline for 6-class classification, tracked with MLflow</li> <li>Trained and optimized classifiers with hyperparameter tuning via Optuna (0.97 AUC)</li> <li>Implemented ML-based region construction and statistical hypothesis testing</li> </ul>         |
|                                                                                                                                                                                          | <b>Tau Anomaly Detection</b><br>Autoencoder   Variational Autoencoder <ul style="list-style-type: none"> <li>Developed an unsupervised ML pipeline for anomaly detection, tracked with W&amp;B</li> <li>Trained and optimized models with Ray Tune (1.46× significance gain)</li> <li>Served reconstruction-error anomaly scores via a REST API (FastAPI, Docker)</li> </ul>                              |
|                                                                                                                                                                                          | <b>Tau Energy Scale</b><br>Mixture Density Network   Neural Networks <ul style="list-style-type: none"> <li>Developed a probabilistic regression model with uncertainty estimation</li> <li>Trained with Huber loss pretraining and NLL fine-tuning for density estimation</li> <li>Evaluated model performance with energy response and resolution histograms</li> </ul>                                 |
| <ul style="list-style-type: none"> <li><a href="#">publication</a></li> </ul>                                                                                                            | <b>Sphalerons and Black Holes</b><br>Convolutional Neural Network   Boosted Decision Trees <ul style="list-style-type: none"> <li>Optimized high-capacity models to classify rare, low-frequency events</li> <li>Built a pipeline encoding raw detector signals as multi-channel images for CNN input</li> <li>Compared high-level features with raw pixel data to maximize signal sensitivity</li> </ul> |
|                                                                                                                                                                                          | <b>Di-Tau Reconstruction</b><br>Software Development   Data Science <ul style="list-style-type: none"> <li>Engineered a C++ analysis framework for large-scale physics reconstruction studies</li> <li>Benchmarked multiple algorithm configurations to optimize reconstruction efficiency</li> <li>Evaluated input variations to establish performance baselines</li> </ul>                              |
|                                                                                                                                                                                          | <b>Elementary Cells</b><br>Software Development   Data Science <ul style="list-style-type: none"> <li>Developed automated C++/Python/LabVIEW software for hardware quality assurance</li> <li>Analyzed array performance via noise, dark count and threshold measurements</li> <li>Explored ML-based classification for detector signal identification</li> </ul>                                         |
| Technical Skills                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Programming Languages &amp; Tools:</b> <ul style="list-style-type: none"> <li>Python, C++, SQL (PostgreSQL), Bash, Linux</li> <li>Git, Jupyter, pytest, uv, conda</li> </ul>          |                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Data Processing &amp; Visualization:</b> <ul style="list-style-type: none"> <li>NumPy, pandas, SciPy</li> <li>Matplotlib, seaborn, Plotly</li> </ul>                                  |                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Machine Learning &amp; Deep Learning:</b> <ul style="list-style-type: none"> <li>scikit-learn, XGBoost</li> <li>PyTorch, PyTorch Lightning</li> <li>Optuna, Ray Tune, SHAP</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Data &amp; ML Infrastructure:</b> <ul style="list-style-type: none"> <li>Docker, GitHub Actions, FastAPI, DVC</li> <li>MLflow, W&amp;B, Hydra</li> </ul>                              |                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Currently Learning:</b> <ul style="list-style-type: none"> <li>PySpark, Databricks, dbt, Azure</li> </ul>                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                           |